MILITARY SPECIFICATION

GENERATOR, SIGNAL: 11726304

1. SCOPE

This specification covers a signal generator used in Test Set, Subassembly MX-8050/TPM-22, part of Radar Test Set An/TPM-22.

2. APPLICABLE DOCUMENTS

The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction; M18 Manufacturing Process and Inspection of

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

D11726304	Signal Generator Assembly

PACKAGING DATA SHEET

11726304	Packaging of Signal Generator Assembly

(Copies of specification, standards, drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The signal generator shall be manufactured in accordance with Drawing D11726304 and drawings pertaining thereto.

3.2 General specification. - Where applicable, the signal generator shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

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3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. - The signal generator shall be mounted on the vibration table in normal service position. The signal generator shall meet the requirements of 3.4 and 3.5 after being subject to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

Frequency Range	Double Amplitude
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle-per-second per second (1 cps/sec). Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. - The signal generator shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11 ± 1 milliseconds.

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3.3.3 Altitude.

3.3.3.1 Operating. - The signal generator shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. - The signal generator shall meet the requirements of 3.4 and 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. - The signal generator shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Nonoperating (storage). - The signal generator shall not suffer permanent damage when subjected to the temperature extremes of -80°F and +185°F, and shall meet the requirements of 3.4 and 3.5 when turned to room ambient.

3.3.5 Humidity-nonoperating. - The signal generator shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram.

3.5 Performance.

3.5.1 Output signals. - When supplied with 115 ± 10 Vac, 400 Hz, between J2-A and B, with OFF X.3 X1 switch in a position other than OFF, and with attenuator control (ATTN) turned fully clockwise, the signal generator shall yield the following outputs.

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OUTPUT	CHARACTERISTICS	TERMINALS
Sine wave	5 V ± 1% peak-to-peak	J1-A & B, P1-D & E
Sawtooth	5 V ± 1% peak-to-peak	P1-H & J
Square wave	5 V ± 1% peak-to-peak, 15 ns rise time.	P1-L & M
Ramp	Amplitude -2.5 ± 1%, 50% duty cycle.	P1-P(-) & R(Ret)
Selectable	Inverse of one of the above, in accordance with position of function selector switch. Variable 0-10V except for ramp which is 0-5V.	P1-B & A(Ret)

Note: All of the above into 50-ohm load.

3.5.2 Attenuation control. - The attenuation control shall serve to vary the amplitude of the selectable output signal of 3.5.1 from 0 V (fully counterclockwise position) to the values indicated in 3.5.1 (fully clockwise position).

3.5.3 Frequency controls. - The X.3 and X1 positions of the OFF X.3 X1 switch, together with the frequency dial and the frequency selector switch shall serve to set the frequency of the output signals of 3.5.1 between 0.005 Hz and 300 kHz.

3.5.4 VERNIER control. - One turn of the VERNIER control shall serve to vary the frequency of the output signals by the amount corresponding to one minor division of the frequency dial.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be listed in MIL-STD-109.

4.2 First article (initial production) approval. - The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of one signal generator manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

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4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the Contracting Officer, inspection lot size, lot information, and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests. - Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and II herein. Examinations and test for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE I - CLASSIFICATION OF DEFECTS

100% INSPECTION

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Electrical	3.4	4.6.2
102.	Output signals	3.5.1	4.6.3.1
103.	Attenuation control	3.5.2	4.6.3.2
104.	Frequency controls	3.5.3	4.6.3.3
105.	VERNIER control	3.5.4	4.6.3.4

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4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. - One signal generator out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table II.

TABLE II - SPECIAL SAMPLING

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable dwg. visual
302.	General specifications	3.2	MIL-F-13926/ MIL-C-46357 visual
303.	Vibration	3.3.1	4.6.1.1
304.	Shock	3.3.2	4.6.1.2
305.	Altitude - operating	3.3.3.1	4.6.1.3
306.	Altitude - nonoperating	3.3.3.2	4.6.1.3
307.	Temperature	3.3.4	4.6.1.4
308.	Humidity - nonoperating	3.3.5	4.6.1.5

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4.4.1 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government-furnished inspection equipment. - Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment.

4.5.2.1 Contractor design. - The contractor shall design and supply inspection equipment compatible with "Test Methods and Procedures" specified in 4.6 of this specification and with the "Component Inspection" procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is considered normally to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to ensure that the test equipment will indicate reliable acceptability of a product that does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of the test equipment.

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4.5.2.2 Test equipment. - In conjunction with 4.5.2.1, the following test equipment shall be used in the performance of the applicable test as specified in 4.6:

TEST EQUIPMENT LIST

ITEM		DESCRIPTION
1.	Power supply	115 ± 10 Vac, 400 Hz, 1A capacity
2.	Ac digital voltmeter	Visual display 5 digits. Full scale range 10.000 volts or greater. Input resistance 1 megohm or greater. Response time 250 ms or less. Accuracy ±0.01% of full scale reading. Meter must register rms voltage.
3.	Electronic counter	Input sensitivity at least 200 mV rms. Registration of six places or more. Range 0 Hz to 300 kHz. Input impedance 1 megohm or greater.
4.	Oscilloscope	Vertical passband DC to 15 MHz. Rise time of scope plus measuring probe 13 ns or less. Sweep period 100 ns per division or less. Input resistance greater than 1 megohm. Input capacitance less than 25 picofarads.
5.	Test harness	Per Figure 1.

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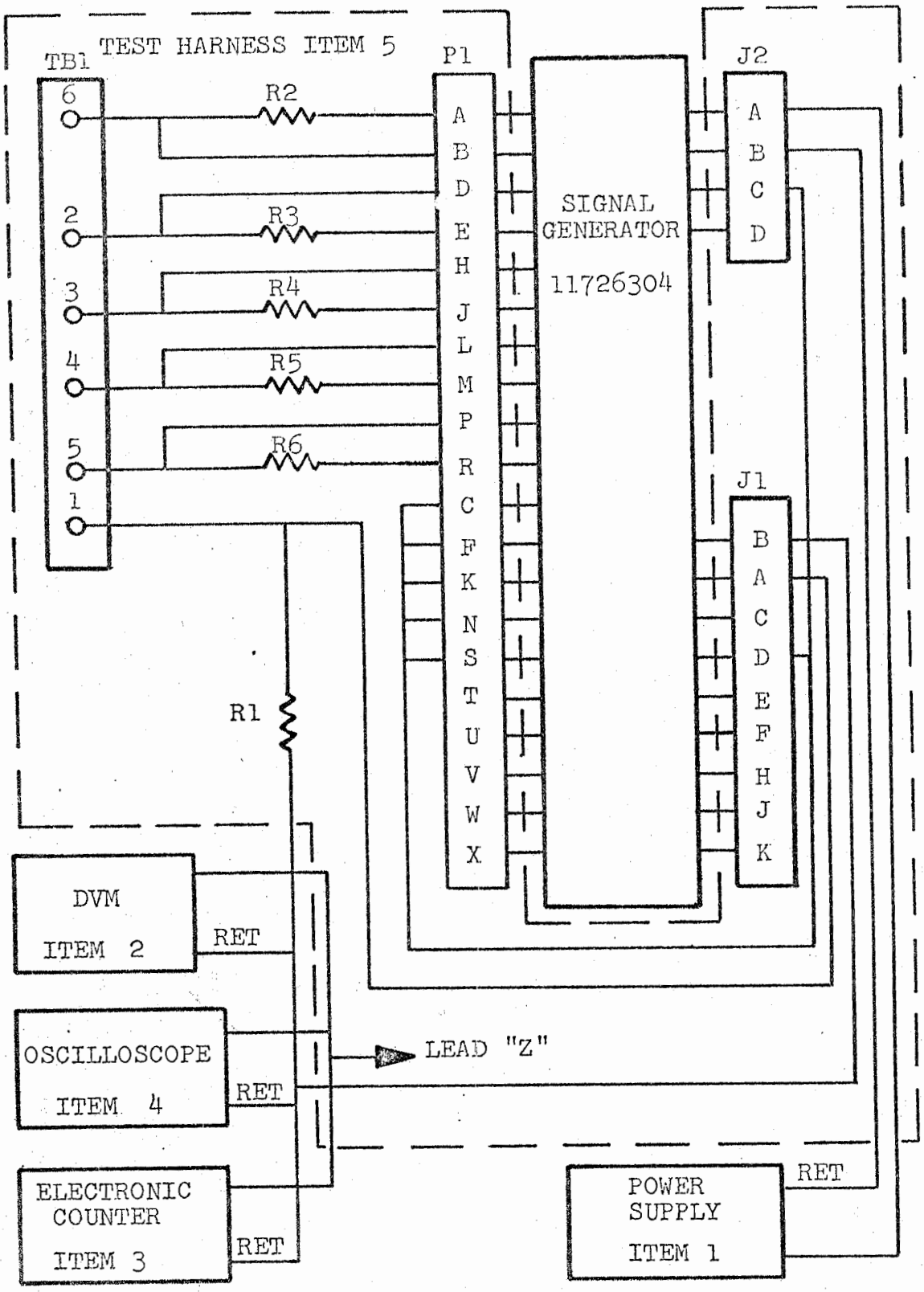


FIGURE 1 - TEST SETUP FOR SIGNAL GENERATOR

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PARTS LIST FOR TEST HARNESS, ITEM 5

DESIGNATION	PART	DESCRIPTION
P1	Connector	MS3116E-12-10P
P2	Connector	MS3116E-8-4S
J1	Connector	Elco 00-8016-020-217-702 or equivalent
R1	Resistor	51 ohms ±5%, 1/4 w
R2	Resistor	Same as R1
R3	Resistor	Same as R1
R4	Resistor	Same as R1
R5	Resistor	Same as R1
R6	Resistor	Same as R1

Wire, Terminal strip, per Figure 1.

4.6 Test methods and procedures.

4.6.1 Environmental. - Environmental test shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.6.1.1 Vibration. - The test shall be performed as specified in Method 514, Procedure IX, Part I, of MIL-STD-810 except that Failure Criteria of Method 514 of MIL-STD-810 do not apply. After vibration, the signal generator shall be tested as given in the sub-paragraphs of 4.6.2 and 4.6.3 for compliance with 3.3.1.

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4.6.1.2 Shock. - This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-810 except that shock pulses shall be 11 ± 1 ms and 15 g's. The G value of each impulse shall be within ±10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5-1/2 milliseconds. After the shock impulses, the signal generator shall be tested as specified in 4.6.2 and 4.6.3 for compliance with 3.3.2.

4.6.1.3 Altitude - non-operating and operating. - This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that in Step 2 the altitude shall be limited to 40,000 ft. (5.5 in Hg). Operation of Step 3 shall be the performance tests of 4.6.3.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and non-operating). - This test shall be performed as specified in Procedure I, Method 501, of MIL-STD-810. The temperature of Step 2 shall be +185°F. The temperature of Step 4 shall be +130°F. In Step 5 the signal generator shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.4.2 Low (operating and non-operating). - This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -85°F maintained for 24 hours. The temperature of Step 4 shall be +32°F. In Steps 5 and 7, the signal generator shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity-nonoperating. - This test shall be performed in accordance with Procedure I, Method 507 of MIL-STD-810 except that in Step 4, the relative humidity shall be maintained at 95%.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

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4.6.3 Performance.

4.6.3.1 Output signals. - Adjust power supply, item 1, to deliver 115 ± 10 Vac and connect to test harness, item 5, as shown in Figure 1. Connect other instruments as shown in Figure 1. Turn OFF X.3 X1 switch on signal generator to OFF and connect signal generator to test harness as shown in Figure 1. Turn attenuator control on signal generator fully clockwise. Turn OFF X.3 X1 switch to X1, frequency selector switch to X1, and frequency dial to 1. Connect lead "Z" of test harness to test harness terminal TB1-1 and observe 1 Hz sine wave on oscilloscope, item 4. Use voltmeter, item 2, to measure amplitude, which should be 1.768 V ± 1% (rms value). Move lead "Z" to test harness terminal TB1-2 and repeat same procedure. Move lead "Z" to test harness terminal TB1-3 and observe sawtooth waveform. Use voltmeter to measure amplitude, which should be 1.443 V ± 1% (rms value). Move lead "Z" to test harness terminal TB1-4 and observe square wave. Use voltmeter to measure amplitude, which should be 2.5 V ± 1% (rms value). Move lead "Z" to test harness terminal TB1-5 and observe ramp waveform described in 5.5.1 on oscilloscope. Use voltmeter to measure amplitude, which should be 1.02 V ± 1% (rms value). Move lead "Z" to test harness terminal TB1-6 and verify that output signal is determined by the position of function selector switch, as specified in 3.5.1. Check all four positions of latter switch.

4.6.3.2 Attenuation control. - With setup existing at end of previous paragrpah, vary setting of attenuation control (ATTN) and observe that amplitude of signal varies smoothly from 0 V (fully counterclockwise) to values indicated in 3.5.1 (fully clockwise).

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4.6.3.3 Frequency controls. - Using electronic counter, item 2, verify that frequency of output signals corresponds to settings of the OFF X.3 XL, frequency selector, and frequency dial of the signal generator. Check each of the sixteen combinations of the two switches, and verify for each combination that the frequency dial varies the frequency over the range indicated by that combination.

4.6.3.4 VERNIER control. - Verify at three frequency settings (low, medium, and high) that VERNIER control varies frequency of signal through an amount equal to approximately one minor division of the frequency dial.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet D11726304. The level of protection shall be as specified in the contract.

6. NOTES

6.1 Intended use. - The signal generator provides distinctive waveforms which can be used in testing and trouble-shooting printed circuit cards.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Applicable packaging data sheet (see 5.1).

(c) Selection of applicable level of preservation packaging and packing.

(d) Level of protection required.

(e) Applicable stock number.

(f) Block of serial numbers to identify each signal generator.

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Preparing Activity

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Project No. 4931-A037

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